Exercise 2

Introduction

The objectives of this exercise are:

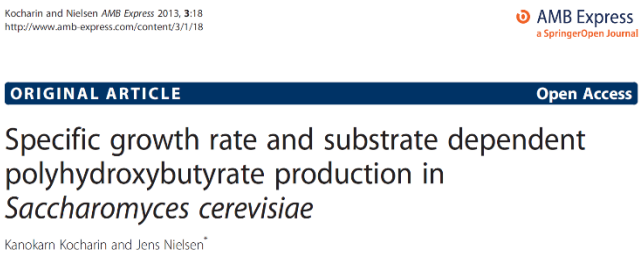
- Formulate an $\boldsymbol{S}$-matrix in MATLAB
- Rearrange an $\boldsymbol{S}$-matrix and note the effect
- Calculate the degrees of freedom and discuss its meaning
- Perform metabolic flux analysis (using a function, but you should know the equation)
- Critically interpret the results from MFA calculations

Report: To help each other, you are encouraged to work in pairs on this exercise. This exercise is not graded and no report needs to be handed in. Meanwhile, the topics being discussed are exam material. **You are advised to work on this Exercise between 21 and 24 September: *after* the lectures on "Construct the S-matrix and flux analysis", and *before* the Live Q&A session** where help will be provided and the solutions to the exercise will be discussed. The solution to this exercise will also be made available on Canvas.

Book contents: This exercise covers chapters 2, 9 and 12 from the Systems Biology textbook: https://doi.org/10.1017/CBO9781139854610.

Exercise files: The current document contains the problems that need to be solved. This exercise is accompanied by a ZIP file containing the necessary MATLAB files (script, function). Extract the ZIP-file to your favorite location and use `exercise2.m` as guidance on what functions you should use to solve the problems. Make your own MATLAB script where you document what command you use to solve each problem.

Background: In this exercise we will construct an $\boldsymbol{S}$-matrix, perform metabolic flux analysis and have a look at rate equations. As an example, we will look at PHB production in yeast. In this paper the authors are looking to improve PHB production through diverting the flux and the effect of different carbon sources.


Kocharin and Nielsen *AMB Express* 2013, **3**:18
http://www.amb-express.com/content/3/1/18

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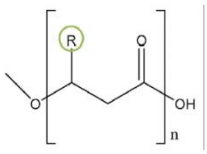
Specific growth rate and substrate dependent polyhydroxybutyrate production in *Saccharomyces cerevisiae*

Kanokarn Kocharin and Jens Nielsen*



R-Group	PHA Name	Short Name
-H	Polyhydroxypropionate	PHP
$-CH_3$	Polyhydroxybutyrate	PHB
$-CH_2CH_3$	Polyhydroxyvalerate	PHV
$-CH_2CH_2CH_3$	Polyhydroxyhexanoate	PHH

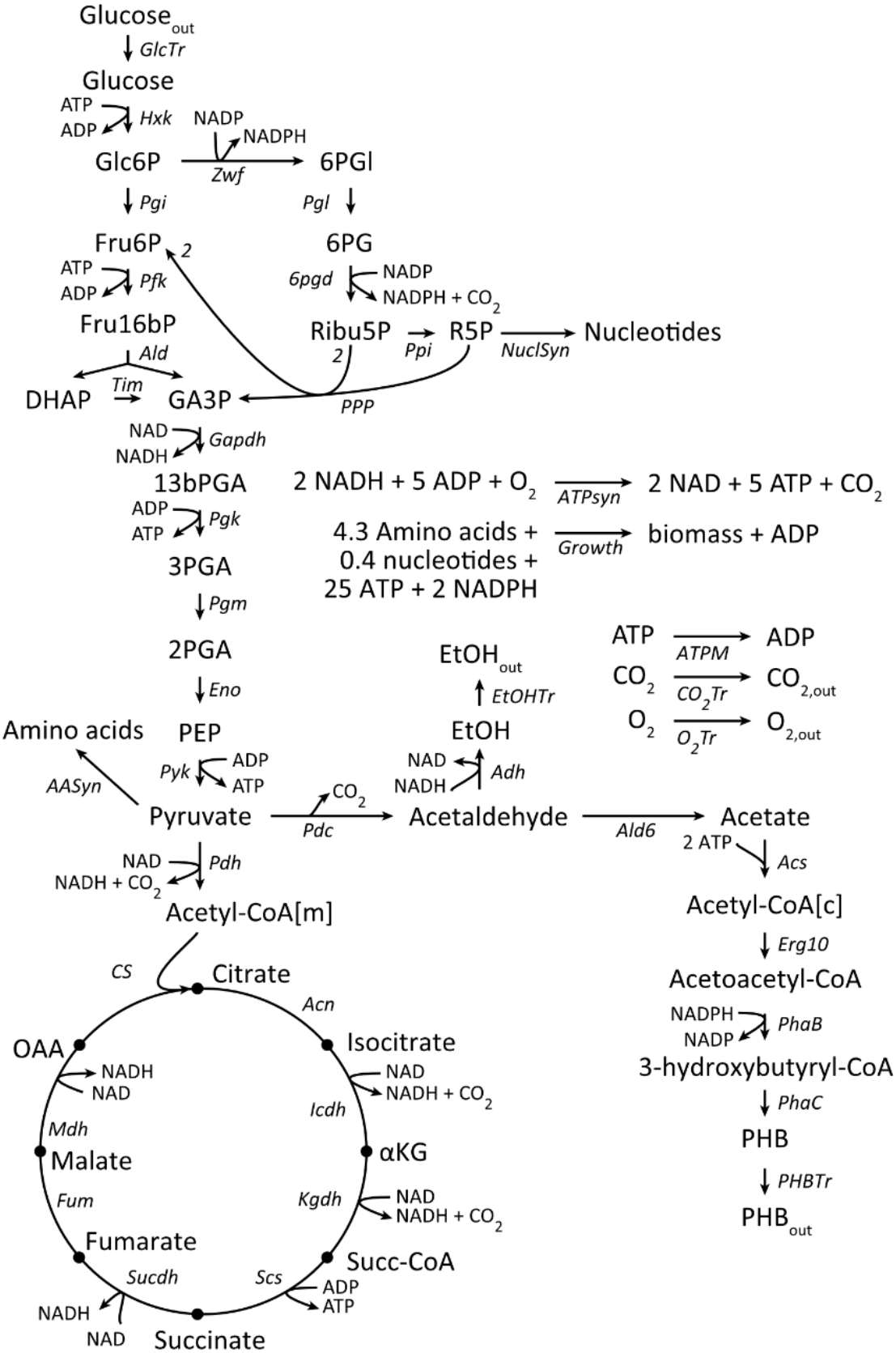
Robin Green, 2010

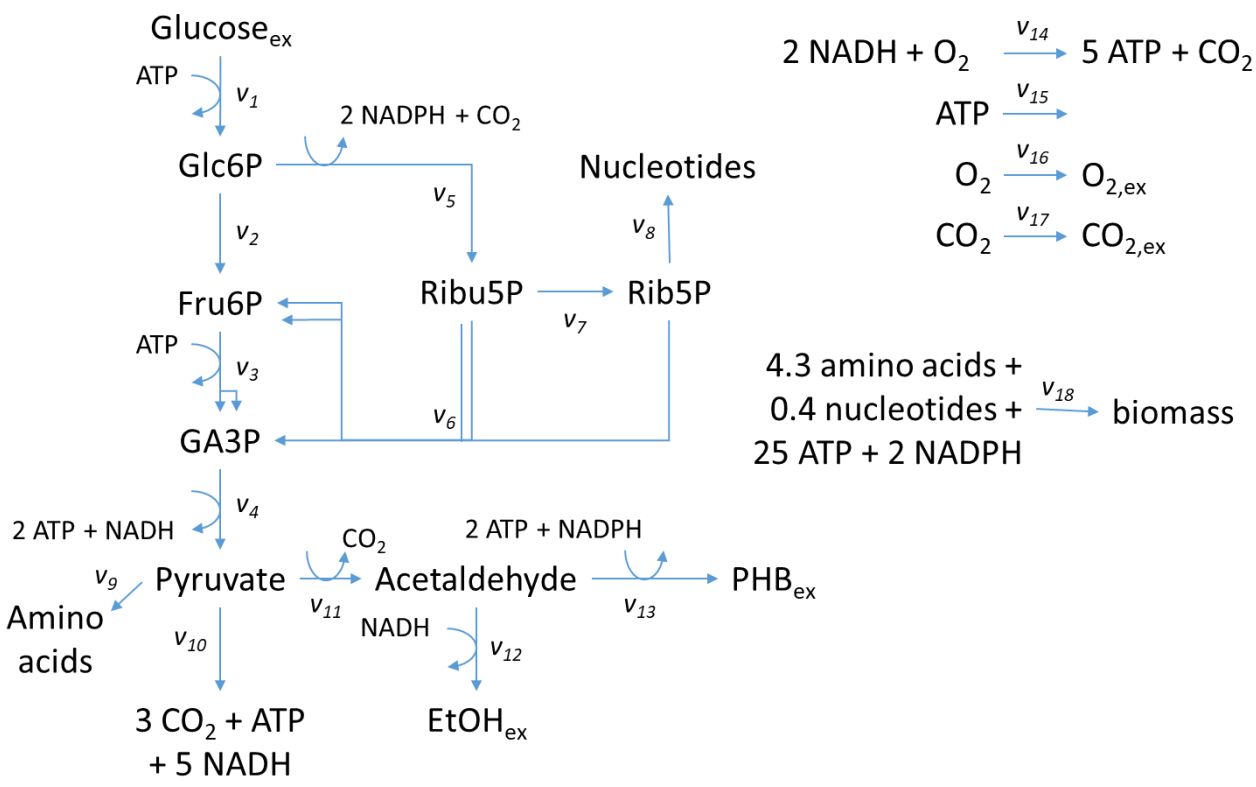
Stoichiometric matrix

On the next is shown a simplified metabolic network including glycolysis, PPP, TCA cycle and PHB production. Note that some reactions are lumped and simplified: AASyn represents amino-acid synthesis; NuclSyn represents nucleotide synthesis; ATPsyn represents ATP synthase. PPP, representing the transaldolase and transketolase reactions, has the reaction equation 2 Ribu5P + R5P → 2 Fru6P + GA3P.

Growth represents the polymerization of biomass precursors to form new biomass. ATPM is an ATP hydrolysis reaction that represents non-growth associated energy costs. Both of these will be discussed in more detail later in the course.

Because we apply the steady-state assumption, we can further simplify this network by lumping linear reactions and removing linear-dependent co-factors. Indicate which simplifications can be made on the network on page 2, and then compare this to the reduced network that is shown on page 3. Did you find all simplifications?





1) The `Exercise2.m` file contains the $\boldsymbol{S}$-matrix, but one column and one row are not filled in yet (indicated with X).
   a) Which reaction and metabolite are these? You can assume that order of reactions is the same as in the figure: first v_1, followed by v_2, v_3 and so forth, with v_{18} last.
   b) Replace every X with the correct stoichiometric coefficients.
   c) Is the $\boldsymbol{S}$-matrix the $\boldsymbol{S}_{tot}$, $\boldsymbol{S}_{in}$ or $\boldsymbol{S}_{ex}$?

2) How many degrees of freedom does the reduced network have, when does this make the model determined and how does this effect the calculation of fluxes?

3) From the reduced network, give two sets of reactions that would match the degrees of freedom and either …
   a) … <u>not</u> be sufficient to calculate all other fluxes:
   b) … <u>be</u> sufficient to calculate all other fluxes (this might be hard to see on paper, but we'll check this in MATLAB later):

4) Assume that we know the ethanol secretion (v_{12}), O_2 consumption (v_{16}), CO_2 excretion (v_{17}) and growth rate (v_{18}), with values 0, -5, 11 and 0.05, respectively. Perform metabolic flux analysis using the code in the MATLAB file.
   a) With what rate is PHB produced in these conditions?
   b) What is the yield of PHB on glucose?

5) The $\boldsymbol{S}$-matrix is defined in a certain order, where the columns and rows correspond to specific reactions and metabolites. However, this order does not have any meaning, we can move columns and rows around while the $\boldsymbol{S}$-matrix still represents the same network and the calculated solutions are still the

same. Do some of these rearrangements in MATLAB to confirm whether this is true, by solving question 4a with the rearranged $\boldsymbol{S}$-matrix.

6) To prevent confusion we will return to using the same $\boldsymbol{S}$-matrix as in question 5 for all the remaining questions. In question 3 you proposed two sets of known reaction rates that either would or would not allow calculation of all fluxes. Check in MATLAB whether your predictions were correct.

7) We want to determine what is the ATP maintenance requirements of the wild-type strain (so there is no PHB production). To do this, we run a continuous cultivation of the yeast in a bioreactor with a culture volume of 500 mL, constant aeration with oxygen and a constant growth rate of 0.05 h^{-1}. The constant growth rate is obtained because there is a constant inflow of fresh medium into the cultivation (and an equal outflow) with the same rate of 0.05 h^{-1}, which means that over 20 hours the culture volumn has been replaced with new culture medium. As an effect, the cells will grow at the same rate as new growth medium is supplied. The glucose concentration in the feed medium is 27 mM, and the amount of cells in the bioreactor, or the biomass concentration, is 2 g dry cell weight (DCW) per liter. From this, you can calculate that the specific glucose uptake rate is 0.675 mmol / gDCW / h. As a bonus question, can you show how this is calculated? [You don't need to be able to answer this question to complete the rest of the exercise]

8) From this cultivation we have also measured the amount of carbon dioxide that is secreted by the cells by measuring the amount coming from the bioreactor, and we have calculated a rate of 3 mmol / gDCW / h. Combine this information with question 6 and calculate the ATP maintenance of the wild-type strain.

9) We also ran an anaerobic cultivation with the same strain, at the same dilution rate, volume and glucose concentration in the feed medium. Compared to in question 6, do you think the amount of cells the bioreactor, or biomass concentration will be higher, lower, or unchanged in comparison with the earlier aerobic cultivation?

10) We measure and calculate a glucose uptake rate of 3.75 mmol/gDCW/h, determine what the ATP maintenance is in these conditions.

11) Now we move on to the engineered strain. Looking at the metabolic network, you can imagine that the PHB production will be highest when there is no growth and no ethanol secretion, this way you are not wasting any carbons! Before doing some experiments, let's first see what the model can tell us about PHB production. Assume the ATP maintenance you calculated in question 7, what are the PHB production rates if the glucose uptake is:
    a) 1 mmol/gDCW/h
    b) 5 mmol/gDCW/h
    c) When the glucose uptake increased five-fold, what did this do to the PHB production rates? Can you explain this?

12) Now have another good look at the reaction rates that you calculated in question 10. Can you spot a problem with the rates in 10a and/or 10b?

13) Now we have actually done an aerobic cultivation with the engineered strain, and we have measured and calculated some rates. The glucose uptake was 4 mmol / gDCW / h, the growth rate 0.3 and no ethanol was secreted.
    a) What was the rate of PHB production?
    b) What was the yield of PHB on glucose? What is the yield if the molecular weight or the number of carbon atoms is taken into account?